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Graded Assignment • 50 min

Due Apr 17, 11:59 PM CEST

Your grade: 80%

Your latest: **80%** • Your highest: **80%** • To pass you need at least 80%. We keep your highest score.

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1. Which of the following do you typically see in ConvNet? — 1 / 1 point

- Multiple FC layers followed by a CONV layer.
- Use of FC layers after flattening the volume to generate output classes.
- Use of multiple POOL layers followed by a CONV layer.
- ConvNet makes exclusive use of CONV layers.

Correct
Yes, FC layers are typically used in the last few layers after flattening the volume to generate the output in classification.

2. In order to be able to build very deep networks, we usually only use pooling layers to downsize the height/width of the activation volumes while convolutions are used with "valid" padding. Otherwise, we would downsize the input of the model too quickly. — 1 / 1 point

- True
- False

Correct
Correct!

3. Training a deeper network (for example, adding additional layers to the network) allows the network to fit more complex functions and thus almost always results in lower training error. For this question, assume we're referring to "plain" networks. — 1 / 1 point

- True
- False

Correct
Correct, Resnets are here to help us train very deep neural networks.

4. The following equation captures the computation in a ResNet block. What goes into the two blanks above? — 1 point

$a^{[l+2]} = g(W^{[l+2]}g(W^{[l+1]}a^{[l]} + b^{[l+1]}) + b^{l+2} + _____) + _____$

- 0 and $z^{[l+1]}$, respectively
- $a^{[l]}$ and 0, respectively
- $z^{[l]}$ and $a^{[l]}$, respectively
- 0 and $a^{[l]}$, respectively

Incorrect
No, $a^{[l]}$ is added before the activation of layer $l+2$

5. Adding a ResNet block to the end of a network makes it deeper. Which of the following is true? — 1 / 1 point

- The number of parameters will decrease due to the shortcut connections.
- The performance of the networks is hurt since we make the network harder to train.
- The performance of the networks doesn't get hurt since the ResNet block can easily approximate the identity function.
- It shifts the behavior of the network to be more like the identity function.

Correct
Yes, as noted in the lectures in a ResNet block the computations are given by $a^{[l+2]} = g(W^{[l+2]}a^{[l+1]} + b^{[l+2]} + a^{[l]})$ thus if $W^{[l+2]}$ and $b^{[l+2]}$ are zero then we get the identity function.

6. For a volume of $125 \times 125 \times 64$ which of the following can be used to reduce this to a $125 \times 125 \times 32$ volume? — 1 / 1 point

- Use a 1×1 convolutional layer with a stride of 2, and 32 filters.
- Use a 1×1 convolutional layer with a stride of 1, and 32 filters.
- Use a POOL layer of size 2×2 with a stride of 2.
- Use a POOL layer of size 2×2 but with a stride of 1.

Correct
Yes, since using 1×1 convolutions is a great way to reduce the depth dimension without affecting the other dimensions.

7. Which of the following are true about the inception Network? (Check all that apply) — 1 / 1 point

- [x] Inception blocks allow the use of a combination of 1x1, 3x3, 5x5 convolutions and pooling by stacking up all the activations resulting from each type of layer.

Correct
Correct. The use of several different types of layers and stacking up the results to get a single volume is at the heart of the inception network.

- [ ] Making an inception network deeper won't hurt the training set performance.
- [ ] Inception blocks allow the use of a combination of 1x1, 3x3, 5x5 convolutions, and pooling by applying one layer after the other.
- [x] One problem with simply stacking up several layers is the computational cost of it.

Correct
Correct. That is why the bottleneck layer is used to reduce the computational cost.

8. When having a small training set to construct a classification model, which of the following is a strategy of transfer learning that you would use to build the model? — 1 point

- Use an open-source network trained in a larger dataset, freeze the softmax layer, and re-train the rest of the layers.
- Use an open-source network trained in a larger dataset freezing the layers and re-train the softmax layer.
- It is always better to train a network from a random initialization to prevent bias in our model.
- Use an open-source network trained in a larger dataset. Use these weights as an initial point for the training of the whole network.

Incorrect
No, the intuition on using transfer learning with small datasets is that the first layers might have learned features that are useful for a new task; but the softmax layer might be significantly different.

9. Which of the following are true about Depth wise-separable convolutions? (Choose all that apply) — 1 / 1 point

- [x] They combine depthwise convolutions with pointwise convolutions.

Correct
Correct, this combination is what we call depth wise separable convolutions.

- [ ] They are just a combination of a normal convolution and a bottleneck layer.
- [ ] The result has always the same number of channels n_c as the input.
- [x] They have a lower computational cost than normal convolutions.

Correct
Yes, as seen in the lectures the use of the depthwise and pointwise convolution reduces the computational cost significantly.

10. Suppose that in a MobileNet v2 Bottleneck block the input volume has shape $64 \times 64 \times 16$. If we use 32 filters for the expansion and 16 filters for the projection. What is the size of the input and output volume of the depthwise convolution, assuming a pad='same'? — 1 / 1 point

- $64 \times 64 \times 32, 64 \times 64 \times 32$
- $64 \times 64 \times 32, 64 \times 64 \times 16$
- $64 \times 64 \times 16, 64 \times 64 \times 32$
- $32 \times 32 \times 32, 32 \times 32 \times 32$

Correct
Correct, the size of the input and output volume of the depthwise convolution is determined by the number of filters in the expansion.